RDguru: A Conversational Intelligent Agent for Rare Diseases

Jian Yang, Liqi Shu, Huilong Duan, and Haomin Li

***Abstract*—Large language models (LLMs) hold significant promise in clinical practice, yet their real-world adoption is constrained by their propensity to produce erroneous and occasionally harmful outputs, particularly in the intricate domain of rare diseases (RDs). This study introduces RDguru, a conversational intelligent agent leveraging the LangChain framework and powered by GPT-3.5-turbo. RDguru offers a comprehensive suite of functionalities, encompassing evidence-traceable knowledge Q&A and professional medical consultations for differential diagnosis (DDX), integrating authoritative knowledge sources and reliable tools. A novel multi-source fusion diagnostic model, rooted in deep Q-network, amalgamates three diagnostic recommendation strategies (GPT-4, PheLR, and phenotype matching) to enhance diagnostic recall during medical consultations. Through tailored tools and advanced algorithms for retrieval-augmented generation, RDguru excels in knowledge Q&A, automated phenotype annotation, and RD DDX. A multi-aspect Q&A analysis demonstrates RDguru outperforms ChatGPT in generating descriptions aligned with authoritative knowledge, quantified by ROUGE scores, GPT-4-based automatic rating, and RAGAs evaluation metrics. Testing on 238 published RD cases reveals that RDguru's top 5 multi-source fusion diagnoses recapture 63.87% of actual diagnoses, marking a 5.47% improvement over the state-of-the-art diagnostic method PheLR. Furthermore, RDguru's consultation strategy proves effective in eliciting diagnostically beneficial phenotypes and refining the prioritization of genuine diagnoses through multi-round phenotype-orient questioning. Evaluations against established benchmarks and real-world patient data demonstrate RDguru's efficacy and reliability, highlighting its potential to enhance clinical decision-making in the realm of RDs.**

***Index Terms*—Conversational AI, deep Q-network, knowledge Q&A, large language model, medical consultation, rare diseases.**

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I. Introduction

RARE diseases (RDs), defined as disorders affecting fewer than 6.5–10 individuals per 10,000, present significant challenges within medical practice due to their individual rarity and complex clinical features [1], [2]. Patients afflicted by these conditions often undergo prolonged diagnostic journeys, often termed the "diagnostic odyssey", owing to the heterogeneous nature and overlapping symptoms of RDs [3], [4]. Despite the establishment of RD-related knowledge bases (KBs) such as Orphanet and OMIM, clinicians frequently encounter difficulties in accessing and utilizing pertinent information, thereby delaying the diagnostic process [5], [6]. However, recent advancements in artificial intelligence (AI), notably the development of large language models, offer promising solutions to these challenges.

Large language models (LLMs), like ChatGPT, have been around for a while and have completely changed many industries, including healthcare and medicine. They are sophisticated neural network-based models that can understand natural language patterns and produce text that is somewhat trustworthy and human-like. Conversational AI provides a more adaptable and natural method of human-machine interaction than conventional keyword query techniques like Orphanet and FindZebra for RD knowledge search [7], [8]. Although general-purpose LLMs exhibit tremendous potential in transforming medical decision support, such as knowledge retrieval, medical diagnosis, and treatment selection, they also harbor inherent limitations [9], [10], [11]. A significant challenge lies in the generation of erroneous information, commonly known as hallucinations, posing a substantial risk, particularly in sensitive domains like healthcare. Moreover, ensuring the credibility of knowledge generated by LLMs remains a critical concern, especially in specialized domains such as RD knowledge retrieval and medical diagnosis. LLMs often struggle to match the credibility of specialized knowledge repositories or purpose-built diagnostic tools, despite their higher operational costs. Addressing these challenges is imperative for leveraging the full capabilities of

LLMs and ensuring their responsible integration into healthcare practices, particularly in the context of RDs.

In efforts to optimize the performance of LLMs within specialized domains, researchers have explored several strategies. One strategy entails fine-tuning LLM using annotated datasets to grasp its knowledge nuances [12], [13]. However, the post-fine-tuned model still essentially relies on word sequence probability prediction and lacks traceable evidence [14]. Another mainstream approach is prompt engineering [15], [16], [17]. This approach aims to inject domain-specialized knowledge or guidance into the context during text generation to steer LLM toward producing more accurate and reliable text, thereby spawning applications where LLMs serve as automated agents. ReAct (Reasoning and Acting) embodies a paradigm wherein LLMs leverage logical reasoning to construct a comprehensive series of actions aligning with desired objectives [18]. In a simple ReAct process, the agent analyzes the steps required for the issue, executes appropriate actions with inferred parameters as input, and then generates a response based on observed information. In the current open-source environment, there are already frameworks like LangChain and AutoGPT that adopt the ReAct approach [19], [20]. Building upon the foundation laid out earlier, the paradigm of LLMs functioning as automated agents enables the integration of LLMs' conversational and reasoning capabilities with reliable and interpretable medical assistance tools, thereby fostering novel AI applications.

Existing diagnosis tools for RDs have predominantly relied on authoritative knowledge repositories, such as Orphanet and Human Phenotype Ontology (HPO) [21], [22], [23], [24], [25]. These tools usually take HPO as input features and design Bayesian semantic analysis algorithms to accommodate the inaccuracies, incompleteness, and noise prevalent in clinical practices. Our previously proposed method, called PheLR, has shown superior performance in the phenotypic diagnosis of RDs and provided excellent clinical interpretability [26], [27]. While general-purpose LLMs exhibit promising performance in phenotypic diagnosis and gene prioritization for RDs, they still lag behind specialized tools in terms of reliability and interpretability [17]. We recently developed another rare disease tool called RDmaster, which is capable of performing differential diagnosis (DDX) for RD patients through phenotype-oriented questioning and answering (Q&A) [28]. However, this system still relies on constrained HPO for user inputs in template-based Q&A, lacking flexibility.

In this study, we introduce RDguru, a conversational intelligent agent tailored for RDs, built on the LangChain framework and incorporating GPT-3.5-turbo as the automated reasoning and summarizing LLM. RDguru's functionalities are centered around knowledge Q&A and medical consultations. We have customized multiple tools for knowledge-enhanced RD knowledge Q&A. With disease entities queried and appropriate processing tools inferred, selected predefined tools will be invoked to retrieve relevant knowledge fragments from online databases such as Orphanet or preloaded Orphadata, ensuring evidence-traceable answer generation. Moreover, RDguru includes a professional tool-enhanced medical consultation process to support interpretable and precise RD diagnosis. This process integrates automated phenotype extraction and context analysis methods for analyzing case text reports, PheLR for the interpretability of phenotypic diagnosis, and RDmaster's phenotype-orient Q&A strategy for differentiating competing diagnoses. A reinforcement learning method based on Deep Q-Networks (DQN), named MixDiagDQN, is proposed for multisource fusion RD diagnosis, aiming to enhance real diagnosis recall by maximizing cumulative rewards [29]. In the evaluation of knowledge Q&A from multiple questioning aspects, RDguru outperformed GPTs in generating descriptions aligned with authoritative knowledge and demonstrated reliable performance within the retrieval-augmented generation assessment (RAGAs) framework [30]. A cohort containing 238 published RD patients was constructed for medical consultation evaluation, in which symptom descriptions of 102 cases were used to evaluate the reliability of RDguru's automated phenotype annotation processes. In the phenotypic diagnosis of this cohort, our proposed MixDiagDQN model outperformed the previously state-of-the-art RD diagnostic method PheLR. Simulated consultations based on these published cases confirmed the RDguru's effectiveness in collecting symptoms beneficial for DDX and elevating the priority of actual diagnoses. Beyond measuring the standalone performance of RDguru, we integrated this conversational AI into an interactive user interface (UI) to facilitate RD-related clinical practice.

II. Materials and Methods

A. LLM-Based Intelligent Agent

Our conversational intelligent agent, RDguru, constructed on the LangChain framework, streamlines the integration of LLMs into applications. The core idea of "agent", as defined in LangChain, involves utilizing a language model as a reasoning engine to take a sequence of pre-defined actions (referred to as "tool") that can handle user messages, reason about appropriate inputs for invoked tools, determine whether the action can be terminated or continue calling other tools, and summarize the output of the last invoked tool as the response. In tools, a sequence of operations is hardcoded (in code). RDguru fulfills two principal functions: providing an interactive Q&A dialogue informed by accurate RD knowledge and conducting DDX in medical consultations upon receiving patient cases using many specialized tools. Considering the substantial token consumption accompanying such LLM-based agent applications, the GPT-3.5-turbo was chosen as the embedded reasoning model to balance reasoning performance and token costs. The technical pathway of RDguru is elucidated in the ensuing subsections.

B. RD Knowledge Q&A

A major challenge in the RD field lies in the limited knowledge among clinicians regarding this subject, and traditional knowledge base queries are often restricted by keyword matching and are not flexible enough. LLMs are capable of engaging in knowledge Q&A in free-text, but often suffer from authenticity hallucinations (such as confusing disease codes) and lack credibility in specialized knowledge. Therefore, a key function of our

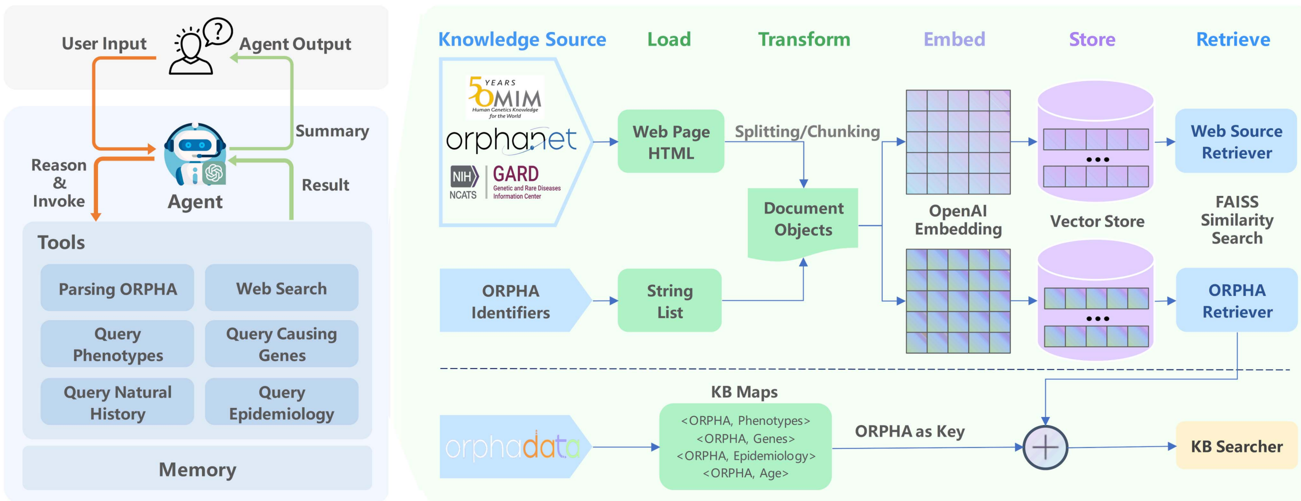

Fig. 1. Retrieval-augmented generation framework for evidence-traceable knowledge Q&A, includes several predefined tools for parsing ORPHA diseases, retrieving online knowledge bases such as Orphanet, OMIM, and GARD, and querying preloaded Orphadata.

intelligent agent is to enhance LLMs through external knowledge sources, enabling them to respond to relevant knowledge questions in natural language, akin to human experts.

Our study introduces a Retrieval-Augmented Generation (RAG) framework, depicted in Fig. 1, which enriches the capabilities of LLMs [31]. It essentially embeds retrieved knowledge into context through prompt engineering, thereby enhancing response accuracy by anchoring the output of the LLM to external knowledge sources such as Orphanet, OMIM, GARD, and Orphadata [5], [6], [32]. Utilizing the LangChain toolkit, we assembled the necessary components for RAG deployment, including document loaders (especially for HTML files of web-based knowledge sources and ORPHA disease name and synonym strings), text splitting and chunking, text embedding (using OpenAI text-embedding-ada-002 model), and vector-based retrievers (using the FAISS (Facebook AI Similarity Search) algorithm). The XML-formatted Orphadata database is parsed and cached as a map within the service memory. In the execution flow, the ORPHA retriever parses RDs that the user inquired about, and then the agent chooses to query the Orphadata map or send a web request. For instance, with Orphanet's web retrieval, after obtaining the Orphanet web response based on parsed ORPHA disease, the response text is chunked into blocks of size 500 (with 50 overlapping between chunks), and embeddings are calculated. FAISS then selects 3 chunks most relevant to the query as the context for enhancing the answer. To cater to the diverse aspects of RD knowledge Q&A, we have tailored specific tools based on the content of different knowledge sources for queries about phenotypes, genetic etiology, and epidemiology, among others.

C. Medical Consultation

Another key function of RDguru is to conduct medical consultations with users to aid in diagnosing cases of suspected RDs. Leveraging LLM as an agent allows RDguru to engage in medical consultations through natural dialogue. In real-world medical diagnostic scenarios, clinicians rely on the patient's chief complaint to make the most probable diagnosis and inquire about symptoms related to potential competing diagnoses to improve diagnostic accuracy. We have divided this process into four sequential sub-tasks for our LLM-based conversational agent and incorporated appropriate methods aimed at enhancing the agent's performance in medical consultations, as depicted in Fig. 2.

1) Phenotype Annotation and Context Analysis: The Human Phenotype Ontology (HPO), extensively used for computational phenotypic analysis and integrating clinical data into translational research, is adopted by most rare disease diagnostic tools to annotate phenotypic abnormalities observed in individuals. Therefore, the first step for the agent is to extract and annotate phenotypes using HPO to better understand the cases described by clinicians and proceed with subsequent diagnosis. The NCBO Annotator developed by BioPortal was integrated to identify HPO terms in case reports [33]. To improve accuracy, we only identify the longest terms from texts and filter out terms that do not belong to the "Phenotypic Abnormality (HP:0000118)" based on HPO. A context algorithm called FastContext, based on the n-trie rule processing engine, is employed to prompt various types of contexts for extracted HPOs, including negation (affirmed or negated), certainty (certain or uncertain), temporality (recent, historical, or hypothetical), and experiencer (patient or non-patient) [34].

2) Symptom Diagnosis: Our previously proposed method, PheLR, a phenotype-driven likelihood ratio (LR) analysis method for RD diagnosis, has demonstrated superior performance compared to most Bayesian-based phenotypic diagnostic approaches [26], [27]. The LR analysis model within PheLR evaluates the contribution of individual phenotypic features (whether present or absent) to candidate diagnoses, employing epidemiological inference for prior probabilities and calculating overall posterior probabilities. However, in a prior study

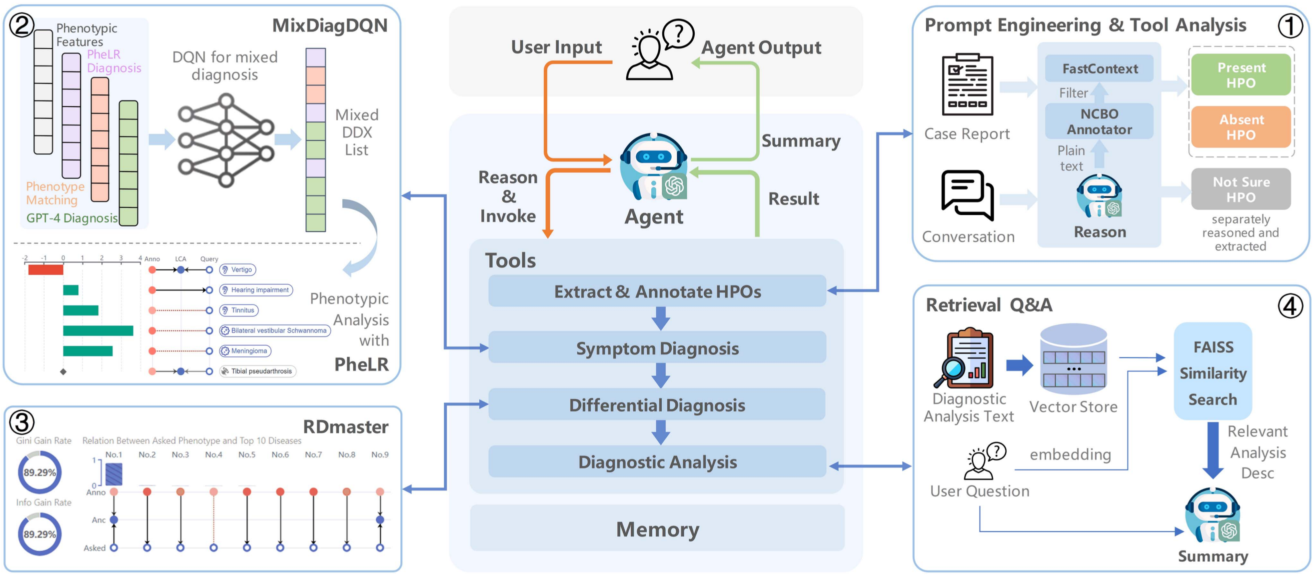

Fig. 2. Retrieval-augmented generation framework for tool-enhanced medical consultations, comprising a continuous workflow consisting of phenotype annotation and context analysis, interpretable symptom diagnosis, differential diagnosis, and diagnostic analysis, each step enhanced by professional tools.

evaluation [28], we observed the unique diagnostic capabilities of LLMs like GPT-4in RD diagnosis. Combining the diagnostic suggestions of GPTs with Bayesian methods such as PheLR has the potential to lead to improved diagnosis. Therefore, in this study, we explore a reinforcement learning algorithm capable of integrating multiple diagnostic approaches (PheLR, GPT-4, and phenotype matching) to improve recall in mixed candidate diagnosis (see next section for details). In mixed diagnosis, the agent also employs the LR analysis model within PheLR to enhance clinical interpretability.

3) Execution of DDX: The extensive overlapping clinical features of RDs often present clinicians with multiple potential diagnoses, necessitating the execution of a DDX to improve accuracy. The RDmaster we developed can handle this situation, wherein the phenotype with the highest expected gain (determined by our proposed Adaptive Information Gain and Gini Index (AIGGI), which is a metric that integrates the overall expected gain of affirming and negating phenotypes) among competing diagnoses is prioritized for inquiry [28]. During each round of medical consultation, based on the reported case, the agent will inquire about three phenotypes belonging to different main classes (e.g., abnormalities of the musculoskeletal system, limbs, or nervous system) with the highest AIGGI score. Subsequently, upon receiving the user's response, it generates a new diagnosis based on the user's assessment of the previously queried phenotypes and provides new clinical features for further DDX.

4) Analysis of Diagnostic Results: A detailed diagnostic analysis report, comprising patient phenotypic feature analysis, PheLR's diagnostic analysis, and RDmaster's DDX details, is constructed as a lengthy text for retrieval Q&A, with a detailed technical process following the RAG framework.

TABLE I
SUMMARY OF COLLECTED CASE REPORTS AND RELATED DATA

Category	Count
Total published case reports	238
Collected patients	
- male	113
- female	119
- unrecorded sex	6
Rare diseases involved	
- total disease number	92
- cases number per disease (mean-median-maximum)	2.6-1-18
HPO terms involved	
- total recorded HPOs over all cases	811
- mean HPO recorded per case	9.1
- mean HPO related to the actual diagnosis per case	5.1
- mean HPO unrelated to the actual diagnosis per case	3.9
- mean HPO excluded per case	1.0

D. A Multi-Source Fusion Diagnosis Model Based on DQN

A previous study indicated a potential complementary diagnostic advantage between KB-based Bayesian methods and LLMs. Building upon this insight, we devised a multi-source fusion diagnosis model called MixDiagDQN, based on Deep Q-networks (DQN) through reinforcement learning [29]. For a given case, this model combines diagnostic results from the PheLR, GPT-4, and common phenotype matching recommendation into a unified DDX list, aiming to achieve state-of-the-art diagnostic performance.

1) Training and Testing Data: We collected data from 238 published patients diagnosed with RDs for testing (summary in Table I). These cases, sourced from PubMed, met strict criteria: each was diagnosed with a known RD, had detailed phenotypic descriptions, included at least two recorded phenotypic features, and its diagnosis has well-documented phenotypic information in Orphadata. We extracted their phenotypic features (present and absent) into HPO terms, categorizing them as follows: (i) "precise": annotated HPO terms of the diagnosed disease; (ii) "imprecise": broader concepts of annotated HPOs; and (iii) "noise": phenotypes unrelated to the diagnosed disease. Due to the rarity of RD cases, we utilized simulated cases for training. We simulated cases based on Orphadata and the phenotypic features of 238 collected patients. For each simulated case, we selected its target disease from 4,257 phenotypically diagnosable RDs based on prevalence, then assigned the number of its phenotypes according to the distribution observed in collected cases, and then allocated its precise (approximately half), imprecise (approximately one-tenth), and noisy (approximately four-tenths) phenotypes according to Orphadata. This process generated a total of 10,000 simulated cases. Randomness was introduced in the simulation to ensure that the distribution and number of phenotypic features in simulated cases closely match those of the 238 collected cases. For each simulated and collected case, three different approaches (PheLR, GPT-4, and phenotype matching) generate 10 ordered recommended RDs, respectively. Phenotype matching prioritizes diseases based on disease-HPO frequency in Orphadata. GPT-4 diagnoses are generated through prompt engineering and natural language processing. The GPT-4 prompt is designed as "Please give a ranked list with ten differential diagnoses (rare disease names only) based on the HPOs provided below," followed by HPO terms. Given that LLMs often confuse disease codes, we employ a method involving regular expression matching and ORPHA retriever to derive the top 10 ORPHA diseases recommended by GPT-4.

2) Policy and Reward Design of MixDiagDQN: We developed a diagnosis manager within a reinforcement learning framework, serving as an agent interacting with the environment represented by a diagnosis simulator. Here, we outline the core components of this task. The diagnostic simulator maintains the patient's phenotypic features, remaining diseases in three candidate DDX lists, and the target ORPHA disease (only in the training phase), and updates the mixed DDX list with a capacity of 10 based on the method chosen by the agent. The diagnosis state s_t consists of three main parts: (i) Patient phenotypic information, represented by the main categories of phenotypic features (23 categories, e.g., the nervous system abnormality, the immune system abnormality, etc.) and the number of present phenotypes. (ii) Diagnostic status information, including top 10 phenotype matching scores, posterior probabilities of the top 10 diseases, and diagnostic uncertainty metrics provided by PheLR. (iii) Fusion diagnosis information, which includes the size of the mixed DDX list and the historical action information of the diagnostic epoch. The state s_t has 87 dimensions, all of which will be normalized for model input. The policy $\pi(a_t|s_t)$ to be learned denotes the probability distribution of the agent taking all possible actions in state s_t. During each round (steps 9–15 of Algorithm 1) in a diagnostic epoch, the agent's action a_t is to select one of the three methods based on its policy $\pi(a_t|s_t)$ and then notify the diagnosis simulator to update its mixed DDX list. A diagnostic epoch concludes when the mixed DDX list size reaches 10.

Algorithm 1: Training of MixDiagDQN.

```
1: Initialization:
2: Initialize Q network and target Q network with random
   weights θ and θ'.
3: Update target network parameters: θ' ← θ.
4: Warm start using an XGBoost model pre-trained on
   training data to pre-fill the experience replay buffer.
5: Training Loop:
6: for each training episode do
7:   for each diagnostic epoch do
8:     Randomly select a training case.
9:     repeat
10:      Observe current state s_t (patient phenotypes,
         mixed diagnosis state, etc).
11:      Select action a_t using ε-greedy policy from
         Q(s, a|θ).
12:      Apply action a_t (choose a diagnostic method).
13:      Receive new state s_{t+1} and reward r_t.
14:      Store experience e(s_t, a_t, r_t, s_{t+1}) in replay buffer.
15:    until mixed DDX list size reaches 10
16:  end for
17:  for each single batch do
18:      Sample random batch from replay buffer.
19:      Update Q network by minimizing L(θ) via root
         mean square propagation.
20:  end for
21:  Update target network parameters: θ' ← θ.
22:  Repeat steps 9–15 to test all testing cases using
     Q(s, a|θ) with ε-greedy set to 0.
23:  Save the best model.
24: end for
```

Rewards play a pivotal role in policy learning. In our task, we incentivize the agent to opt for the appropriate diagnostic method to improve the recall of the target disease within the mixed DDX list. The reward policy is designed as follows: Each candidate DDX list resembles a stack, where the most probable disease resides at the top. In a diagnostic round of steps 9–15 of Algorithm 1, the agent selects a method, which involves popping the top disease off the corresponding DDX stack and then enqueueing the popped disease into the mixed DDX list. After popping a disease, we evaluate whether the minimum distance of the target disease to the top of each of the three DDX stacks has decreased. Ideally, we expect the method chosen by the agent to be the one where the target disease is closest to the top among all candidate methods, and the minimum distance will decrease in this case. If it does, we reward the agent's action a_t with a value of 1; conversely, we assign a reward of -1. Furthermore, if the popped disease matches the target disease, the reward is increased to 2. The goal of this reward policy is to incorporate target diseases that appear in different recommendation methods into the final DDX list as quickly as possible.

3) Training With Deep Q-Network: Q-learning is a reinforcement learning algorithm used to estimate the cumulative expected reward, denoted as $Q(s_t, a_t)$, when taking action a_t in a given state s_t, with the goal of maximizing cumulative rewards over time. Deep Q-Network (DQN) is a method that combines deep learning techniques with Q-learning to learn policies in environments with large state spaces. In this task, we employed a multi-layer perceptron (MLP) which takes s_t as input and output estimates of $Q(s_t, a_t)$, and then selects action a_t based on the ϵ-greedy strategy. The MLP structure comprises a three-layer neural network with a hidden layer of 512 units, using the rectified linear unit (ReLU) as the activation function for the hidden layer and the sigmoid activation function for the output layer. Two crucial DQN tricks, namely target network usage and experience replay, are applied in this task [35]. According to the Bellman equation, the expected Q-value can be calculated as follows:

$$Q(s_t, a_t|\theta) = r_t + \gamma \max_{a_{t+1}} Q^*(s_{t+1}, a_{t+1}|\theta') \tag{1}$$

θ and θ' represent the parameters of the training network and target network, respectively. γ denotes a discount rate (set to 0.9). r_t is the reward obtained by acting a_t in state s_t.

In each training episode (steps 6–24 of Algorithm 1), 100 diagnostic epochs are conducted, in which all experiences (denoted as $e(s_t, a_t, r_t, s_{t+1})$ for a diagnostic round) are updated into the experience replay pool. The experience replay buffer size is set to 10,000, allowing approximately 10% to 20% of experiences to be updated in each training episode. Since the diagnostic performance of PheLR significantly outperforms the other two methods, the ϵ value of ϵ-greedy strategy is set to 0.6 (step 11 of Algorithm 1) during the training phase to ensure effective action space exploration. After 100 diagnostic epochs, we sample from the experience replay pool for batch training. The batch size is set to 64. We employ the mean squared error loss function and choose root mean square propagation (RMSprop) as the optimizer, with a learning rate of 0.03 for batch training. The loss function can be expressed as:

$$\begin{aligned} L(\theta) = \mathbb{E}_{(s,a,r,s')}\Big[\Big(r + \gamma \max_{a'} Q^*(s', a'|\theta') \\ - Q(s, a|\theta)\Big)^2\Big] \end{aligned} \tag{2}$$

At the end of each training episode, the parameters of the target network are updated, followed by the execution of the testing process, and the parameters of the best-performing model are saved. Additionally, we trained a simple XGBoost model on the training data for warm starting, which was utilized to pre-fill the experience replay buffer before the agent's exploration-exploitation. This trick allows the DQN to learn effective strategies initially and reduces the number of training episodes needed.

III. Results

A conversational intelligent agent named RDguru for rare diseases was designed and developed as an online platform in this study (http://rdguru.nbscn.org/). RDguru is hosted on a cloud server, ensuring reliability and ease of access. A brief demonstration video is accessible on the RDguru platform homepage. RDguru integrates two crucial scenarios, knowledge retrieval, and medical consultation, in the form of a chatbot, providing clinicians with an easy-to-access expert agent for rare diseases.

A. RD Knowledge Q&A of RDguru

1) Perform Enhanced Knowledge Q&a With RDguru Platform: The UI of RDguru for knowledge Q&A is illustrated in Fig. 3(a). When users express concern about a rare disease, the built-in LLM in RDguru will recognize the user's intent and invoke the appropriate tool for processing. RDguru utilizes authoritative website information or KBs to gather and condense relevant details, providing feedback to the user and offering links to the original knowledge sources for comprehensive information verification. The ORPHA retriever, utilizing embedding similarity computation, accommodates deviations in the expression of disease entities queried by users, e.g., as depicted in Fig. 3(a), the user's non-standard description of "blackfan diamond anemia" is correctly aligned with "Diamond-Blackfan anemia" in Orphadata. In multi-turn conversations, the memory module enables the agent to continue responding to disease entities from the historical records. The tab accompanying the agent's response provides knowledge retrieved from external sources, enhancing the credibility of the agent's response. For the knowledge Q&A agent, the available tools are designed based on the diverse content of KBs or web resources. For inquiries that surpass the capabilities of its built-in tools, RDguru utilizes the embedded chat model to address them.

2) Quality Assessment of Knowledge Q&A: Evaluating text generation tasks such as knowledge Q&A is a complex endeavor. In this study, we introduced two pipelines for evaluating the quality of knowledge Q&A. The first involves assessing the similarity of generated text to authoritative knowledge descriptions relevant to the query. This includes three evaluation methods: (i) The widely used automated metric ROUGE, employed for measuring text similarity and evaluating information overlap, specifically utilizes ROUGE-1, ROUGE-2, and ROUGE-L scores, covering unigrams, bigrams, and longest common subsequence overlap, respectively [36]. (ii) A custom noun phrase-based accuracy and recall evaluation (NP-ARE) approach, utilizing the en-core-web-sm model provided by spaCy to extract noun phrases from the text, incorporating stop words and lemmatization to evaluate the noun phrases-based similarity of texts. (iii) A third LLM (GPT-4) agent serves as human evaluation to compare and score the original knowledge statement with the generated response, conducted using prompt engineering (see details in the appendix). The second pipeline involves the RAGAs framework specifically designed to evaluate RAG's performance [30], which provides various metrics for both retrieval and generation aspects, including precision and recall of the context (retrieved from the external knowledge source for answering questions), faithfulness, answer relevancy, and answer correctness. All metrics are scaled to the range [0, 1], with higher values indicating a better performance.

To refine the evaluation results, we conducted knowledge Q&A across 8 questioning aspects (100 questions per aspect), utilizing a total of 23 question templates filled with disease names or synonyms. The GPT-3.5 and GPT-4 were introduced

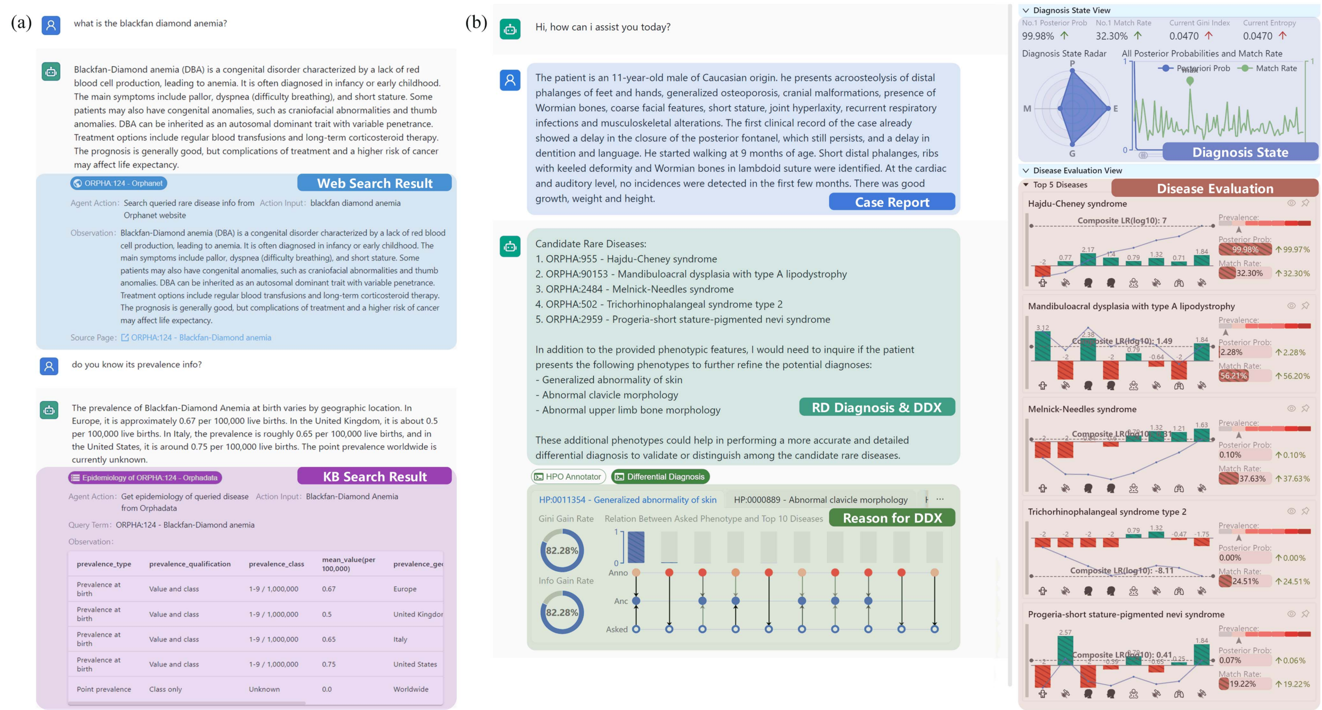

Fig. 3. User interface (UI) of RDguru. (a) UI for evidence-traceable knowledge Q&A, additionally displaying parsed ORPHA diseases and information queried from online knowledge bases or Orphadata. (b) UI for tool-enhanced interpretable medical consultations, additionally showcasing phenotypic evaluation for the top five diagnoses supported by PheLR, differential diagnosis explanations supported by RDmaster, and real-time diagnostic states such as diagnostic uncertainty.

TABLE II
EVALUATION RESULTS (MEAN ± STANDARD DEVIATION) OF RDGURU AND GPTS FROM THE RAGAS FRAMEWORK

	Context Precision	Context Recall	Faithfulness	Answer Relevancy	Answer Correctness	Answer Correctness of GPT-3.5	Answer Correctness of GPT-4
Symptom	1.00 ± 0.00	1.00 ± 0.00	0.83 ± 0.32	0.90 ± 0.25	0.85 ± 0.26	0.57 ± 0.21	0.54 ± 0.21
Causing Gene	1.00 ± 0.00	1.00 ± 0.00	0.60 ± 0.38	0.94 ± 0.19	0.75 ± 0.23	0.39 ± 0.21	0.48 ± 0.26
Epidemiology	1.00 ± 0.00	1.00 ± 0.00	0.55 ± 0.33	0.58 ± 0.48	0.59 ± 0.21	0.44 ± 0.23	0.40 ± 0.20
Natural History	1.00 ± 0.00	1.00 ± 0.00	0.82 ± 0.33	0.33 ± 0.45	0.84 ± 0.26	0.41 ± 0.25	0.39 ± 0.23
Inheritance Mode	1.00 ± 0.00	1.00 ± 0.00	0.91 ± 0.27	0.56 ± 0.48	0.90 ± 0.23	0.46 ± 0.27	0.49 ± 0.28
Diagnostic Methods	0.64 ± 0.35	0.65 ± 0.41	0.49 ± 0.36	0.91 ± 0.21	0.75 ± 0.20	0.58 ± 0.19	0.60 ± 0.21
Differential Diagnosis	0.63 ± 0.38	0.71 ± 0.43	0.43 ± 0.36	0.86 ± 0.24	0.59 ± 0.30	0.48 ± 0.23	0.39 ± 0.19
Management & Treatment	0.72 ± 0.28	0.78 ± 0.32	0.57 ± 0.33	0.72 ± 0.40	0.74 ± 0.22	0.56 ± 0.19	0.58 ± 0.18
Overall	0.87 ± 0.26	0.89 ± 0.28	0.65 ± 0.37	0.73 ± 0.41	0.75 ± 0.26	0.49 ± 0.23	0.48 ± 0.24

as competing methods. For the first evaluation pipeline, GPTs were able to complete all assessments. For the second evaluation pipeline, since the native GPTs did not have context retrieved for enhancing answers, they could only be assessed for answer correctness.

For the first evaluation pipeline, with the original knowledge descriptions as the target text, the average precision, recall, and F1 score of 100 responses from the three methods in each aspect are illustrated in Fig. 4. The results suggest that overall, RDguru outperforms GPTs in knowledge Q&A related to RDs. In terms of recall in ROUGE-1 and NP-APE, RDguru shows a significant advantage for symptoms and natural history Q&A, with slight advantages in other aspects. Regarding precision in ROUGE scores and NP-APE, RDguru significantly surpasses GPTs in all aspects, indicating that RDguru provides more concise answers with higher consistency with the original knowledge descriptions. In GPT-4-Eval, acting as a simulated human assessment, RDguru performs comparably with GPTs in differential diagnosis Q&A, with slight advantages in management and treatment and diagnostic methods Q&A, and substantial superiority in other aspects over GPTs.

For the second evaluation pipeline, the results in Table II indicate that RDguru significantly outperforms GPTs in answer correctness. For the first five rows of Orphadata-related questioning aspects, both context precision and recall reached the upper limit, suggesting that RDguru retrieves highly accurate contexts as long as correctly identifying the ORPHA disease for Orphadata-related queries. For Orphanet web-related questioning aspects (rows 6 to 8), there is a noticeable decline in both context precision and recall. This drop may be attributed to the chunking strategy (selecting 3 chunks as context, each of size 500, a total context size of approximately 1500), which

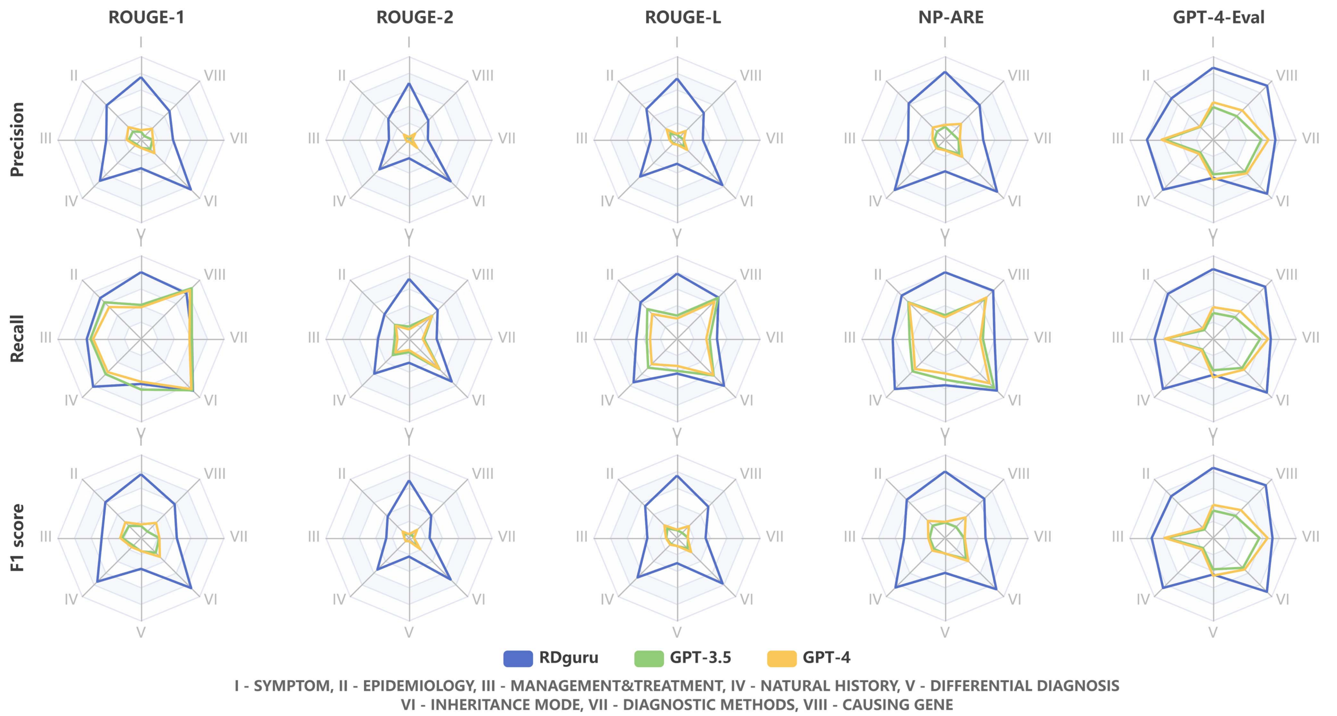

Fig. 4. Quality assessment results of knowledge answering of RDguru and GPTs from 8 questioning aspects, calculating precision, recall, and F1 score using ROUGE scores, custom NP-ARE, and GPT-4 automatic evaluation, with authoritative knowledge descriptions as the target. Radar charts indicate scores ranging from 0 to 1.

may include extraneous information unrelated to the query, and the FAISS's retrieval of query-relevant chunks is not extremely precise. The faithfulness and answer relevancy of RDguru are acceptable. Notably, all metrics, including GPTs' answer correctness, show large standard deviations, indicating some instability in response generation. Even within the same questioning aspect, variability can be significant due to differences in the diseases queried. Overall, RDguru's RAG framework for knowledge Q&A performed well in the RAGAs evaluation.

3) Tool Invocation and ORPHA Retrieve: We further analyzed the details of tool invocation by the agent. Out of a total of 800 Q&A cases across 8 questioning aspects, the agent failed to invoke any tools in 49 (6.13%) cases but opted to provide a direct response. For the majority of this situation, simply reformulating the question could resolve the issue. When the agent did invoke a tool, there was a 1.33% (10 out of 751) chance of failing to identify the correct ORPHA disease, and a 1.33% (10 out of 751) chance of not invoking the most appropriate tool to address the question. Overall, the invocation of tools and parsing of ORPHA are sufficiently robust.

B. Medical Consultation of RDguru

1) Perform Medical Consultation With RDguru Platform: RDguru's UI for medical consultation is shown in Fig. 3(b). In this scenario, users can provide detailed descriptions of clinical cases. and then RDguru will automatically extract the mentioned phenotypic features into HPO terms, suggest potential diagnoses (limited to 5) based on MixDiagDQN, and investigate phenotypes for further DDX. Alongside the agent's response, two accompanying tabs offer the extracted HPOs and explanations from RDmaster regarding the DDX, specifically illustrating relationship graphs between the queried phenotypes and the top 10 candidate diseases. The diagnostic state view primarily consists of a radar chart indicating the posterior probability and phenotypic match ratio of the top-ranked disease, along with the entropy and Gini index derived from the posterior probability distribution of the top 10 candidate diseases. A larger area on this radar chart corresponds to a higher confidence level in the diagnosis. The disease evaluation view provides a detailed depiction of the quantitative contribution of individual phenotypic features to various diagnoses, as measured by LR scores, thereby enhancing diagnostic interpretability. RDguru typically proceeds with querying whether the current patient exhibits specific phenotypes after presenting possible candidate diagnoses. Upon user response, the patient's phenotypic features will be updated, initiating a new round of DDX. Note that the current release of RDguru is capable of diagnosing 4,257 RDs with recorded phenotypic information in Orphadata. The subsequent diagnostic evaluations utilize this set of 4,257 RDs as candidate diseases.

2) Evaluation of Phenotype Annotation: We quantitatively evaluated the performance of RDguru's phenotype annotation pipeline using a dataset consisting of 102 out of 238 published RD patients who described their cases in textual format (while others mostly in tabular form). We manually annotated the present phenotypes (totaling 1,018 HPOs) and absent phenotypes (totaling 97 HPOs) in these 102 case descriptions as the gold standard. In addition to the phenotype annotation pipeline (NCBO&FastContext) used in this study, two other pipelines

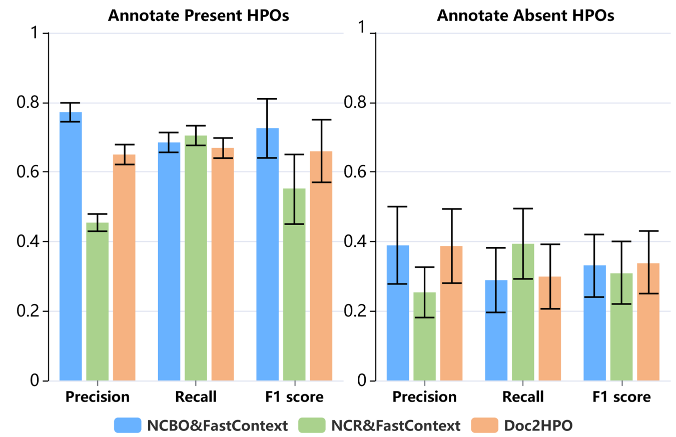

Fig. 5. The precision, recall, and F1 score of phenotypic extraction and context analysis using different approaches. The error bars indicate the 95% confidence intervals. Results of phenotype parsing for patient's present phenotypes on the left subplot, and parsing results of patient's absent phenotypes on the right subplot.

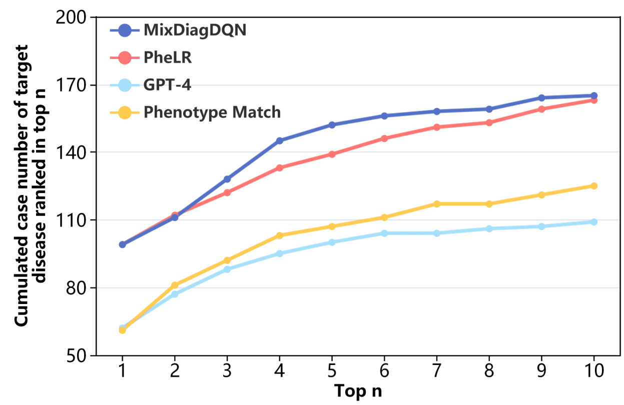

Fig. 6. Accumulated number of target diseases ranked in the top 10 in phenotypic diagnosis evaluation across 238 RD cases using MixDiagDQN and the individual methods (PheLR, GPT-4, and phenotype matching) it integrates.

were introduced as competing methods: Neural Concept Recognizer (NCR), which lacks negated annotation and thus combines with FastContext, and Doc2HPO, which integrates the NegEx algorithm for negated annotation [37], [38], [39].

The evaluation results are depicted in Fig. 5. For annotating present phenotypes, NCBO&FastContext (precision 0.77, 95% Confidence Interval (CI): 0.74–0.80; recall 0.68, 95% CI: 0.66–0.71, F1 score 0.73, 95% CI: 0.64–0.81) outperforms Doc2HPO (precision 0.65, 95% CI: 0.62–0.68; recall 0.67, 95% CI: 0.64–0.70, F1 score 0.66, 95% CI: 0.57–0.75) comprehensively. Although NCR&FastContext achieves the highest recall (0.70, 95% CI: 0.68–0.73), its excessive extraction of HPOs leads to poor precision (0.45, 95% CI: 0.43–0.48). For annotating absent phenotypes, NCBO&FastContext performs comparably to Doc2HPO, and NCR still results in high recall and low precision due to over-extraction of HPOs.

3) Evaluation of RD Recommendation Using MixDiagDQN: The effectiveness of the multi-source fusion diagnosis model MixDiagDQN was evaluated using 238 published RD cases. All present phenotypes in each test case were reported for diagnosis. The final mixed diagnosis list generated by MixDiagDQN contains no duplicate diseases.

As depicted in Fig. 6, MixDiagDQN achieves recall rates of 69.33% (165 out of 238 cases) for the true diagnoses in the top 10 positions, 63.87% (152 out of 238 cases) for the top 5, and 41.60% (99 out of 238 cases) for the top 1. The standalone PheLR achieves recall rates of 41.60%, 58.40%, and 68.49% for the top 1, top 5, and top 10 positions, respectively. Similarly, GPT-4 demonstrates recall rates of 26.05%, 42.02%, and 45.80%, while phenotype matching shows recall rates of 25.63%, 44.96%, and 52.52% for the same positions. Compared to individual diagnostic methods, MixDiagDQN's multi-source fusion diagnosis strategy demonstrates superior performance, particularly in the recall at the top 5 positions, achieving a 5.47 percentage point improvement over the state-of-the-art Bayesian method PheLR for RD diagnosis. Considering the breadth of 4,257 candidate RDs covered, a 63.87% recall in the top 5 is an extremely high diagnostic performance.

We further investigate MixDiagDQN's policy in detail. Diagnosed using MixDiagDQN, the source methods of the top 10 diagnoses for each test case and the distribution of real diagnoses are illustrated in Fig. 7. In terms of diagnostic method sources analysis, PheLR dominates the recommendations in the first position, likely because it provided significantly more correct diagnoses in the first position than the other two methods. For the second position, MixDiagDQN primarily chose results from phenotype matching, with some from PheLR. This likely occurs because phenotype matching showed performance second only to PheLR in the first and second positions, as shown in Fig. 6. It can be observed in Fig. 8 that there is a significant overlap in diagnoses between phenotype matching and PheLR, likely because both are KB-based methods. Given PheLR's superior diagnostic performance, MixDiagDQN rarely selects phenotype matching in subsequent positions. Conversely, although GPT-4's diagnostic performance is slightly inferior to phenotype matching, it has fewer overlapping diagnoses with PheLR, which benefits diagnosis complementary. Therefore, MixDiagDQN incorporates some recommendations from GPT-4 in the third, fourth, and lower positions, which contribute to the significant improvement in recall rate within the top five positions.

4) Evaluation of Phenotype-Oriented Multi-Round Q&A for DDX: We assessed the diagnostic performance of RDguru within the phenotype-oriented multi-round Q&A. Among the 238 collected cases, half of their phenotypic features were randomly selected as an initial report, followed by fixed rounds (set to 10 rounds) of phenotype-oriented Q&A. Each patient underwent three experiments to accommodate variations in the initial report. In response to phenotypes queried by the agent, users affirmed the present phenotypes of the patient, negated the excluded phenotypes, and responded "uncertain" if the queried phenotype was not mentioned in the case report.

Out of a total of 714 dialogues, 422 (59.10%) captured valuable information, where users affirmed or negated certain phenotypes. Among these, 91 (21.56%) dialogues showed an

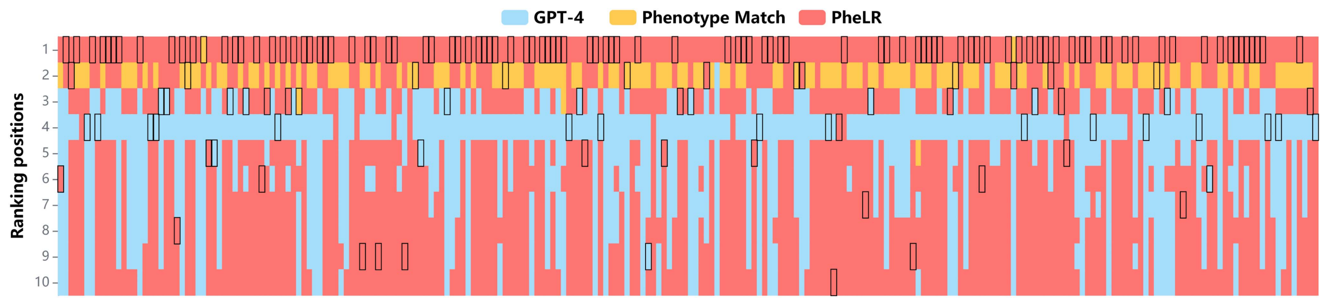

Fig. 7. The performance of MixDiagDQN in 238 published cases. The background color shows the source of the recommended diagnosis. The solid black line marks the true diagnosis of the test case.

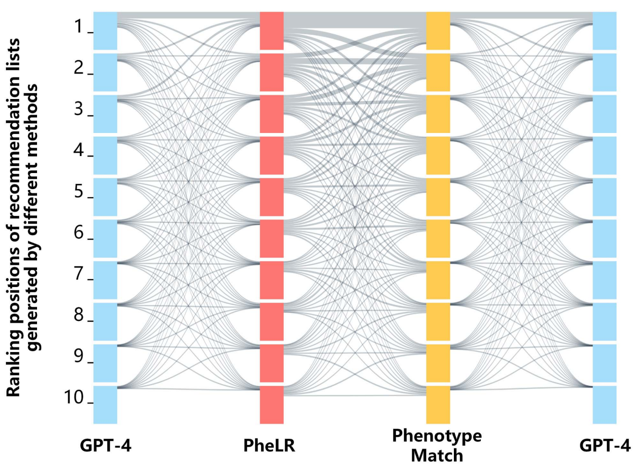

Fig. 8. The overlap of the top ten recommended diseases among PheLR, GPT-4, and phenotype matching in testing 238 published cases, displayed using a Sankey diagram. Each item size in the positions of methods is 238, and the width of connecting lines between two items represents the corresponding number of disease overlaps.

improvement in the ranking of real diagnoses, with an increase of 23 (5.45%) instances where the real diagnosis ranked first. Out of the 627 phenotypes affirmed by users, 467 (74.48%) were associated with their respective real diagnoses, aiding in the diagnostic process. Out of the 233 symptoms negated by users, 71 (30.47%) were related to their real diagnoses, potentially affecting its ranking - a common occurrence as patients typically do not exhibit all symptoms related to real diagnoses, while excluding phenotypes related to other diseases also aids in diagnosis.

During a dialogue diagnosis, RDguru queries 3 phenotypes per round, totaling 30 phenotypes after ten rounds. 59.10% of RDguru's dialogue test captured valuable information, which may seem inefficient. However, RDguru can perform phenotypic diagnosis for 4,257 RDs, associated with 9,087 different phenotypes forming the candidate set for the phenotype-oriented Q&A. Given that each published case has an average of 9.1 recorded phenotypes, and assuming five remaining phenotypes for Q&A, the probability of randomly selecting 30 phenotypes from 9,087 candidates without replacement and capturing at least one of the remaining phenotypes is less than 2%. Therefore, achieving a capture rate of 59.10% demonstrates RDguru's exceptional effectiveness.

IV. Discussion

While LLM-based chatbots like ChatGPT have outperformed Isabel Pro in diagnosing common diseases [40], [41], their effectiveness in RD consultations is limited due to the underrepresentation of RDs in large text corpora and their inherent complexity [42]. Existing text corpora for training LLMs often suffer from inconsistencies, incompleteness, or information bias without having undergone authoritative validation, resulting in LLMs being criticized for spreading false information with persuasive rhetoric and artificial hallucinations (e.g., confusing disease coding). Such a shortage is particularly crucial in the realm of RDs, as confused patients or clinicians attempt to associate symptoms with diagnoses and treatments, whereas thousands of RDs are collectively related to about ten thousand phenotypes (an average of a dozen or so per disease). For RDs, clinicians are more likely to expect diagnostic assistance and seek AI tools than common diseases. Validating that an LLM-based intelligent agent can provide credible, reliable, and visualized diagnostic assistance in the field of RDs is a great start for the entire medical field. This is where RAG technology comes into play, enhancing LLMs by augmenting their ability to generate responses with information retrieved from a wide range of reliable and up-to-date sources. Besides authoritative KBs, RDguru integrates various proven tools for different purposes to support medical consultations. Thus, it can go beyond these tools and provide better and more intelligent services by fusing them. LLM enhanced by our RAG frameworks results in significant improvements in both knowledge Q&A and medical consultation tasks by achieving evidence traceability, answer credibility, and diagnostic accuracy.

There is currently a plethora of other research on domain-specialized LLMs for RDs. The majority of these studies fundamentally involve fine-tuning open-source LLMs or conducting zero-shot or few-shot prompt engineering. Li et al. fine-tuned Llama using medical dialogue datasets and integrated an information retrieval process to obtain and utilize real-time information from Wikipedia [12]. Tang et al. utilized LLMs as collaborators for zero-shot role-playing in multi-turn discussions within medical reasoning scenarios [15]. Compared to general LLM-based medical applications, our study focuses on RDs to adequately address its clinical challenges, especially diagnostic difficulties. There have also been endeavors in RD-related LLM-based applications. Shyr et al. utilized zero-shot and few-shot learning with LLMs to identify rare disease and phenotype entities, while Wang et al. fine-tuned Llama2 to align given symptoms with HPO IDs [13], [16]. Kafkas et al. employed zero-shot and one-shot (chain-of-thought) methodologies within prompts to present symptoms and candidate genes, facilitating LLMs in prioritizing causative genes [17]. Although this approach of prompting LLMs for medical diagnosis achieves better results than native LLMs, it still lacks sufficient efficiency and applicability, especially in terms of clinical interpretability, which is precisely the focus of RDguru.

Several unique methodological contributions that set RDguru apart from previous works include (i) Advanced conversational agent for RDs. Our tailored RAG frameworks greatly enhance its professionalism, accuracy, and credibility. (ii) Innovative multi-source fusion diagnosis. This is a pioneering research exploring the integration of multiple diagnostic paradigms to improve diagnostic accuracy. (iii) Enhanced conversational interface for RDs. Its natural conversational interface offers greater flexibility and ease of use compared to traditional knowledge query and diagnostic tools. For fusion diagnosis, previous research compared KB-based diagnostic tools and GPTs for generating DDX lists for RDs, revealing that while standalone GPT-4 performs well, likely due to its inclusion of RD training corpus, its combination with PheLR significantly improves diagnostic accuracy. [28]. For instance, GPTs exhibit significantly higher diagnostic efficacy for cases with a higher prevalence than cases with a lower prevalence, while KB-based tools did not show a statistically significant difference in diagnosing diseases with different prevalence. GPTs perform better for cases with fewer reported phenotypes, whereas PheLR shows the opposite trend. Given the limited research on combining different diagnostic paradigms, we present a multi-source fusion diagnosis model with a DQN-based reinforcement learning strategy, demonstrating superior performance over standalone optimal diagnostic method PheLR and offering a promising direction for integrating different algorithms for the same purpose to achieve superior performance.

Evaluating generative AI like RDguru for knowledge Q&A using standardized machine learning (ML) error metrics such as ROUGE scores (precision, recall, and F1) may not fully capture the nuances required in the medical domain. The RAGAs framework is still fundamentally based on LLM-driven automated evaluation. Human evaluation metrics, such as domain-specific metrics (like relative factual correctness, completeness, conciseness, and potential harm) and domain-agnostic metrics (like readability scores), provide a more comprehensive assessment [43]. However, RDguru's broad coverage of over 7,000 RDs makes finding comprehensive human evaluators impractical. Therefore, metrics like ROUGE scores and NP-ARE, complemented by third-party GPT-4 evaluations, provide a feasible alternative. The ML error metrics such as precision and recall align closely with human evaluation metrics. For instance, the precision score provided by ROUGE can reflect the relative conciseness of the generated text in terms of words or phrases, and recall scores can reflect relative completeness. Meanwhile, RDguru's feature of professional KB augmentation and evidence traceability reduces potential harm, and the integration of advanced LLMs such as GPT-3.5-turbo ensures the readability of its generated responses. It is difficult to define a clear threshold for what can be considered a successful metric, as it varies depending on the type of evaluation metrics and questioning aspects. RDguru's overall higher scores than the introduced baselines can still be considered valuable. Although human evaluation remains ideal, our approach balances feasibility and comprehensive evaluation.

For MixDiagDQN training, We simulated cases to generate patients similar to actual patients due to the scarcity of real RD cases, as training with only hundreds of collected cases is challenging. As we mentioned, we only simulated the distribution and number of phenotypic features based on the actual cases for training, while the actual cases were used for testing. The target diagnosis for simulated cases was randomly selected from all 4,257 phenotypically diagnosable RDs. The 10,000 simulated cases used for training involve a total of 936 RDs, while the 238 actual cases involve only 92 RDs. Therefore, testing on published patient cases will not introduce bias or overfitting into the diagnostic results. Furthermore, components from our previous studies in RDguru's medical consultation agent, such as PheLR and RDmaster [26], [28], are not machine learning models and do not learn from or memorize the 238 collected cases, thus eliminating concerns about data leakage.

RDguru's performance relies on the reasoning capability of embedded LLM, robustness of internal prompts, quality of the supporting knowledge, and integrated tools and models. As a result, despite being evaluated using template-based questions and online cases, significant deviations are not anticipated in actual applications. RDguru's modular design enables it to adapt to evolving LLMs. As LLMs advance, RDguru is expected to enhance its capabilities, e.g., more accurate tool invocation and more human-like responses. Designed specifically for RDs, RDguru stays focused on its domain by politely guiding users when their inputs are off-topic.

Regarding concerns about ablation studies for RDguru, it's important to note that, in knowledge Q&A, RDguru effectively serves as GPT-3.5 combined with our RAG framework, and its performance is compared with that of GPTs, essentially functioning as an ablation test of our RAG framework for knowledge Q&A. For medical consultation, RDguru integrates multiple modules into a continuous execution flow, where each tool complements others in symptom extraction, diagnosis, and multi-turn dialogue diagnosis. Therefore, isolated ablation studies are not feasible due to the interdependent nature of these modules.

There are several limitations to this study. Firstly, applications built using the LangChain framework require pre-defined and coded actions to meet specific design needs and thus lack suitable tools for unforeseen scenarios and issues. While RDguru currently has numerous tools, any future addition of new requirements will require further manual development, potentially incurring significant costs. Secondly, while this study demonstrates significant advantages in phenotypic diagnosis through published case testing, the diagnosis of RDs remains a formidable challenge. Given that the majority of RDs are genetic disorders and leveraging variant data to interpret RDs is imperative for advancing diagnostic efficacy, future endeavors can focus on parsing and diagnosing patients with sequencing data via conversational AI applications. Thirdly, although we have exhaustively evaluated the performance of RDguru, an in-depth evaluation of its true clinical performance is needed in the future to ensure the reliability and effectiveness of RDguru in real-world medical settings. Lastly, RDguru is specifically tailored for RDs and does not address non-RD cases or broader disease categories, which limits its applicability in general clinical practice. We hope to conduct further research in the future to expand its applicability and better serve clinical experts.

V. Conclusion

In conclusion, we constructed an LLM-based conversational intelligent agent for RDs, which can provide evidence-traceable knowledge retrieval and reliable differential diagnosis. A novel multi-source fusion diagnosis model based on DQN integrates the recommendation result of LLM with the current advantageous algorithms and improves the diagnostic effect of the system as a whole. This study presents not only a technical achievement but also a promising step forward in transforming healthcare through biomedical informatics and emerging AI techniques.

Appendix

Here we present the prompt for assessing the quality of knowledge Q&A using GPT-4. SYSTEM PROMPT: "You are an AI evaluator specializing in assessing the quality of disease knowledge Q&A. Rate the information precision (IP) and information recall (IR) of TEXTs for the given QUESTION, based solely on the given KB DESCRIPTION to the given QUESTION obtained from knowledge bases. IP - int — Score from 1 to 5. IP = 1 means the TEXT is almost inaccurate/irrelevant for KB DESCRIPTION, IP = 3 means about half of the information in TEXT matches for KB DESCRIPTION, and IP = 5 means the information in TEXT is completely accurate for KB DESCRIPTION. IR - int — Score from 1 to 5. IR=1 means that TEXT has almost no recall of the effective information of KB DESCRIPTION, IP = 3 means that TEXT has recalled about half of the effective information of KB DESCRIPTION, and IP = 5 means that TEXT has recalled almost all the effective information of KB DESCRIPTION. Directly give the rates. Example Answer: "TEXT1 - IP: 2, IR: 3; TEXT2 - IP: 3, IR: 4; TEXT3 - IP: 3, IR: 2"". USER PROMPT: "QUESTION: {}; KB DESCRIPTION: {}; TEXT1: {}; TEXT2: {}; TEXT3: {}"

Conflict of Interests

All authors have reported that they have no relationships relevant to the contents of this paper to disclose.

Data Availability

This study utilized data from 238 rare disease patients published online in PubMed. Detailed information about the test patients, including their PMIDs and identifiers, extracted HPO terms, and text reports, are available at https://github.com/1678492347/rdguru-test. The source code of MixDiagDQN is available at https://github.com/1678492347/mix-diag-dqn.

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